Examiner only

**2.** The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

| | | | | | | |
|---|---|---|---|---|---|---|
| radio waves | microwaves | infra-red | visible light | ultraviolet | X-rays | gamma rays |

(a) **Tick** (✓) the **three** correct statements. [3]

The frequency of em waves decreases from radio waves to gamma rays. ☐

All em waves travel at the same speed in a vacuum. ☐

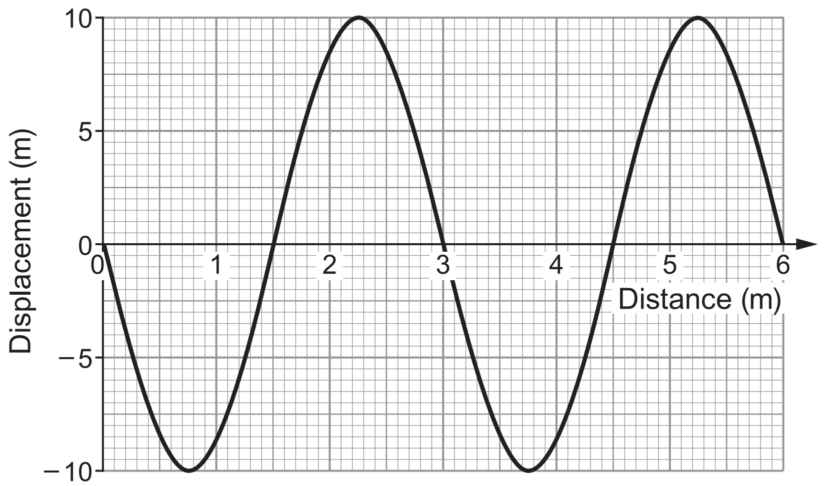
Gamma rays can be emitted from radioactive materials. ☐

The wavelength of em waves decreases from radio waves to gamma rays. ☐

Radio waves, microwaves, infra-red and visible light are all ionising radiations. ☐

All em waves are longitudinal. ☐

(b) The graph below shows a wave diagram.




**Use the following values** to answer the questions below. You may use each value once, more than once or not at all.

| | | | | |
|---|---|---|---|---|
| 2 | 3 | 5 | 6 | 10 |

(i) I. State the amplitude of the wave. .......................................... m [1]

II. State the number of complete waves shown. .......................................... waves [1]

III. State the wavelength of the wave. .......................................... m [1]

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(ii) I. Complete the following sentences by underlining the correct phrase in each bracket. [2]

The wave has a frequency of (**7.5 Hz / 7.5 cm/s / 7.5 cm**).

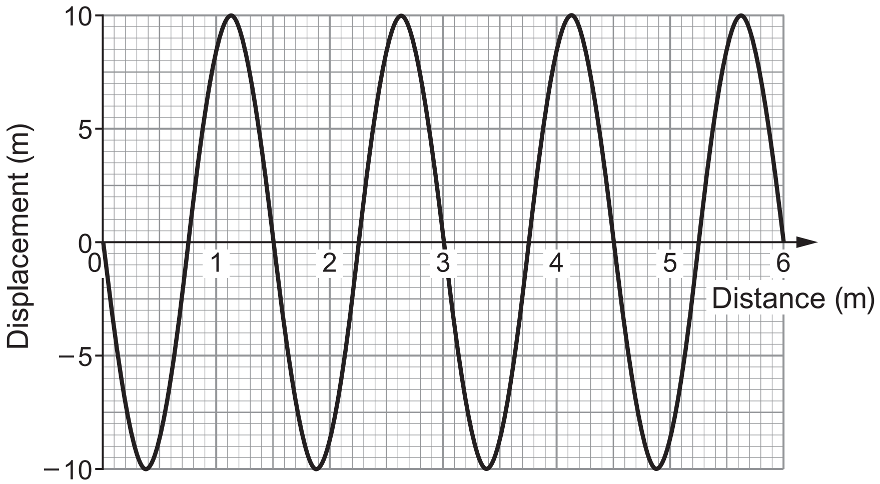
This means 7.5 waves pass a point each (**second / minute / hour**).

II. Use information from parts (b)(i) and (ii) to calculate the wave speed. Use the equation:

$$\text{wave speed} = \text{wavelength} \times \text{frequency}$$ [2]

Wave speed = .............................. m/s

(c) The speed of the wave remains constant but its **frequency is doubled**. A student draws a diagram of the expected wave. It is shown below.




Explain whether you agree with the student's diagram of the wave. [2]

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